



Computer
Science

CSC380: Principles of Data Science

Introduction and Course Overview

Xinchen Yu

Course Staff

Instructors:

- Xinchun Yu, Ph.D.
 - Office hours: Thursdays, 12:00pm-2:00pm (priority window 1:00-2:00pm)
- Naren Doraiswamy
 - Office hours: TBD

Course TAs:

- Ruoshan Lan, Praneeth Myneedu
- Office hours: TBD

Today's Plan

- Data Science Introduction
- Course and syllabus overview

What is “Data Science”?

My Definition: *The process of using data to (1) answer questions, (2) extract knowledge, and (3) predict future outcomes.*

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Examples:

- Do people in college towns tend to buy more laptops than people in other areas?

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Examples:

- Do people in college towns tend to buy more laptops than people in other areas?
- Find out top-10 sales categories for each age group.
- Summarize product reviews with respect to product quality.

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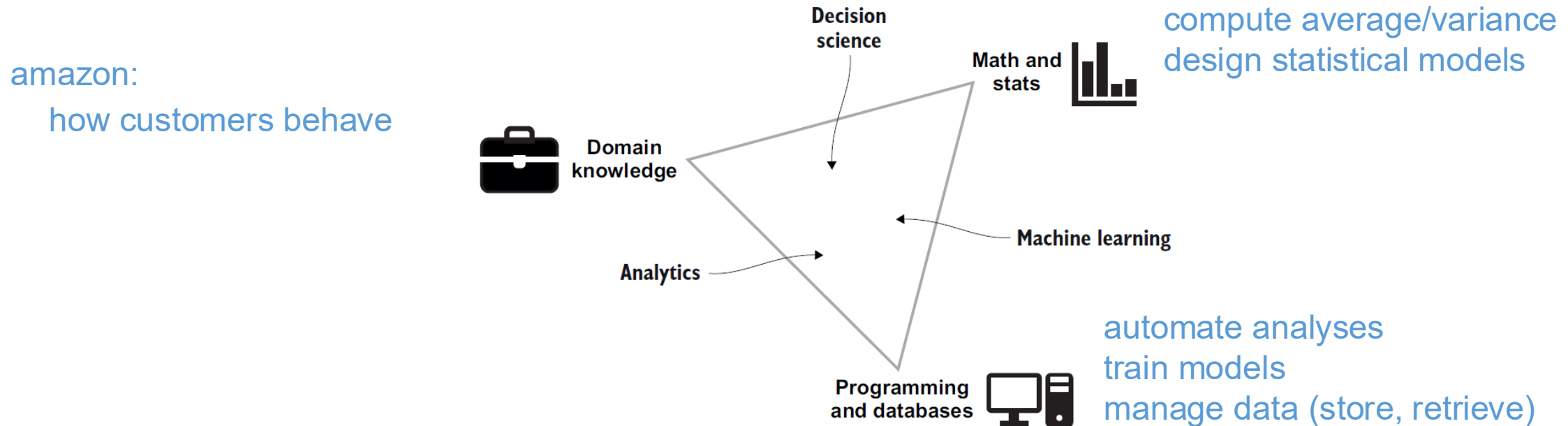


Examples:

- Do people in college towns tend to buy more laptops than people in other areas?
- Find out top-10 sales categories for each age group.
- Summarize product reviews with respect to product quality.
- If we recommend pens to users from college town, how much will it increase our revenue?

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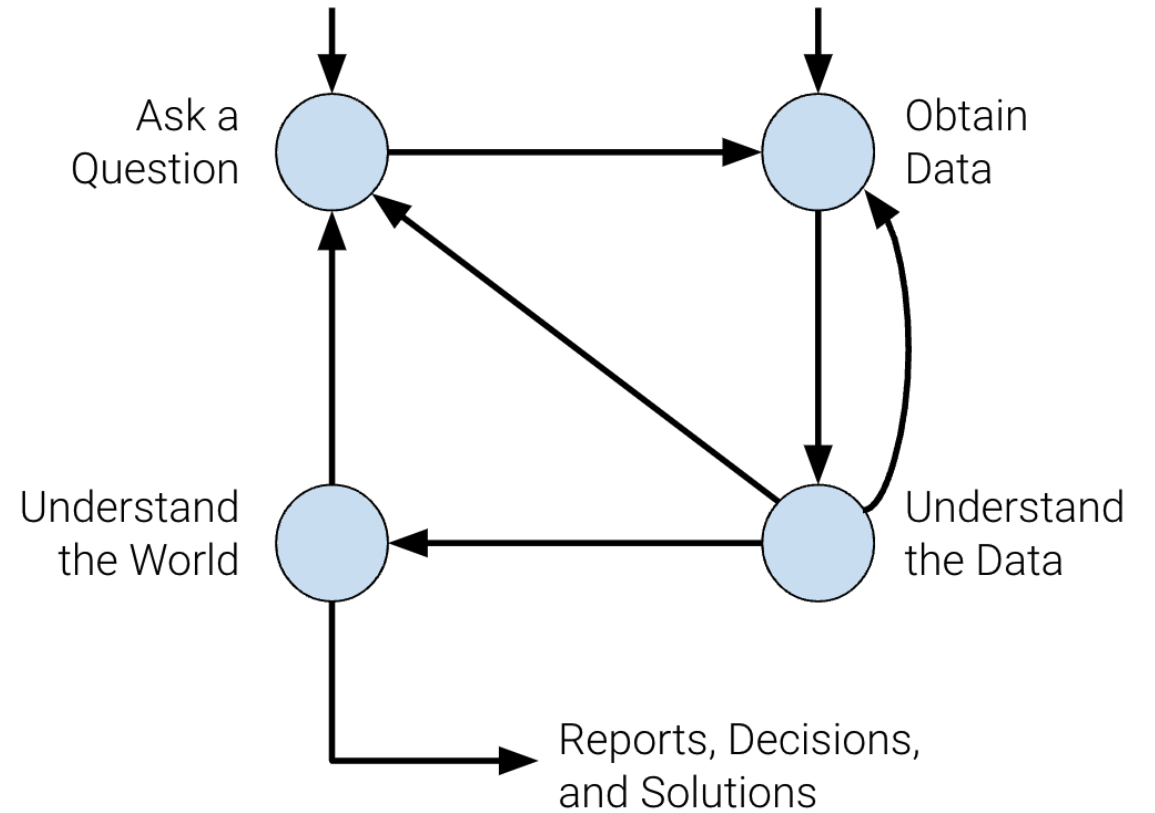


[Source: [Robinson, E. and Nolis, J.](#)]

Data Science Is:

- **Interdisciplinary:** Combines tools and techniques from Math / Statistics / CS
- **Exploratory:** Understanding data requires creative exploration and visualization
- **Applied Statistics & Probability** + extra stuff to handle, process, and visualize data

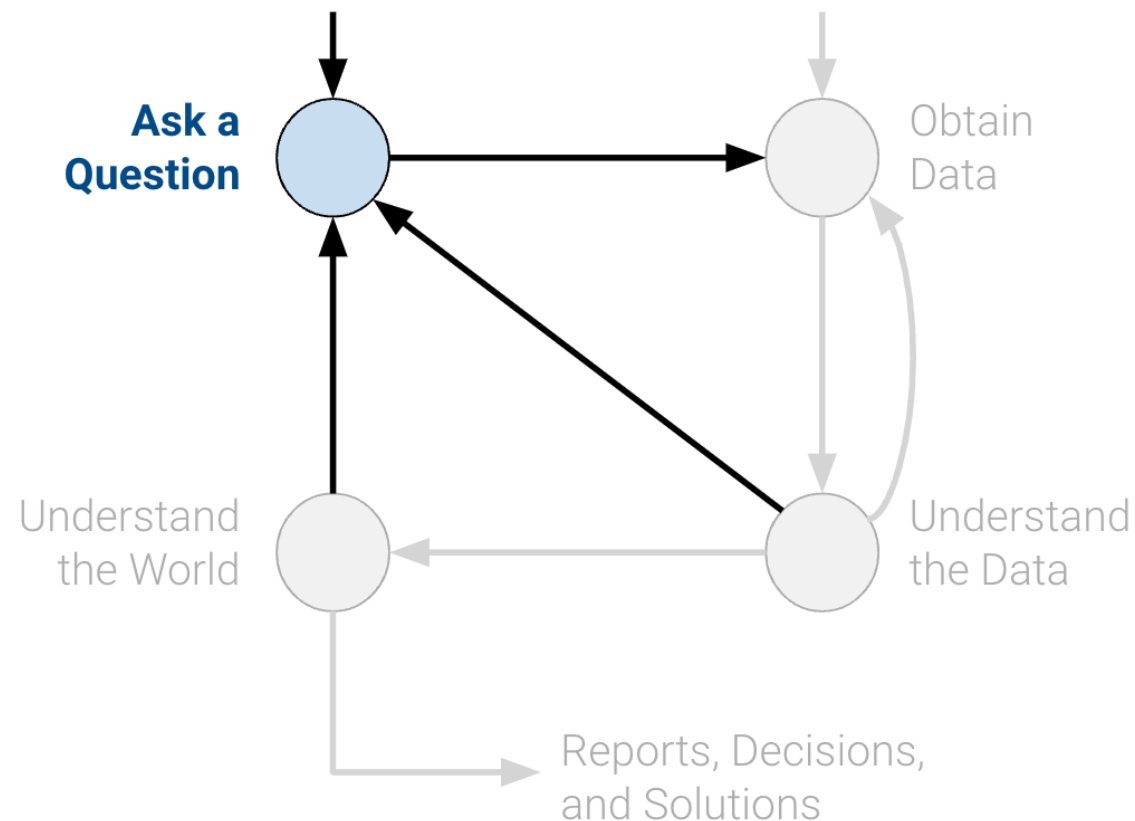
The data science life cycle



Data 100, UC Berkeley

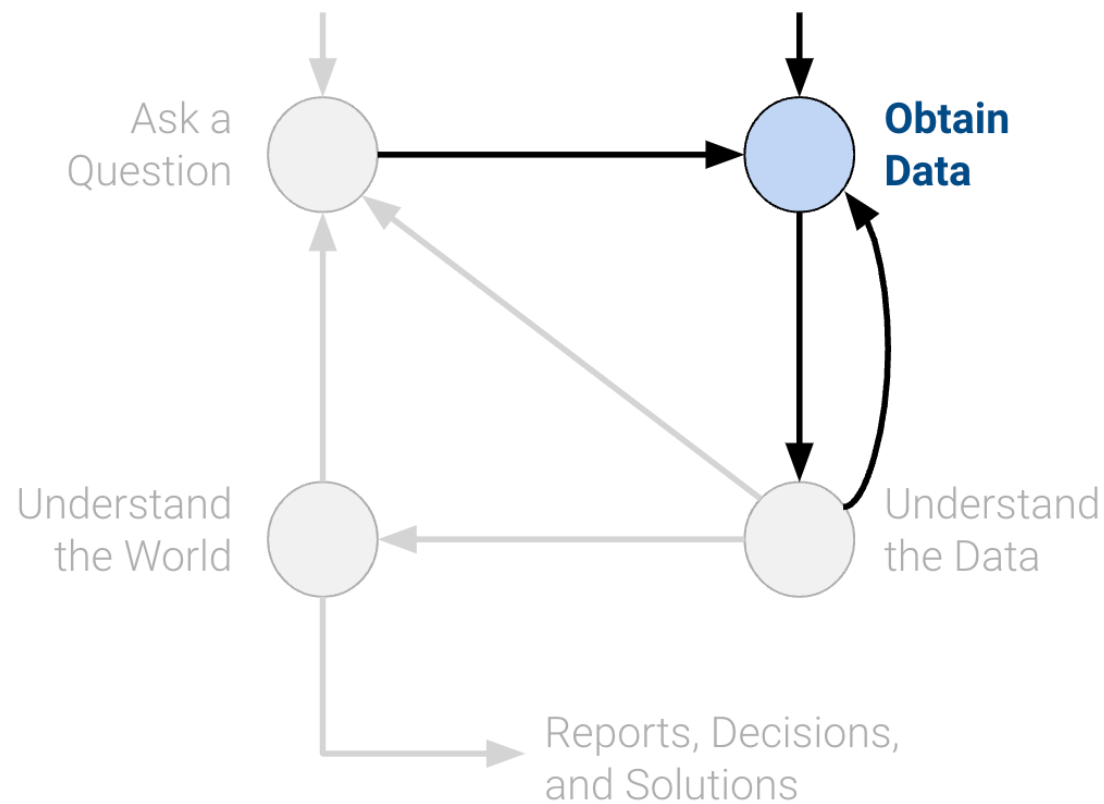
The data science life cycle

- What do we want to know?
 - What is the age distribution of lead actors and actresses in highly rated movies?
 - What does the collaboration (co-cast) network of actors and actresses look like?
- What problems are we solving?
- What hypotheses are we testing?
- What are our metrics?



The data science life cycle

- What data do we have and what data do we need?
- How will we sample more data?
- Is our data representative of the population we want to study?



An example

- The IMDb movie database: Movies

imdb.com/title/tt0133093/?ref_=chttp_i_16

The Matrix

1999 · R · 2h 16m

IMDb RATING **8.7/10** 2M

YOUR

Details [Back to top](#) [Edit](#)

Release date [March 31, 1999 \(United States\)](#)

Countries of origin [United States](#) · [Australia](#)

Official sites [HBO Max \(United States\)](#) · [Official Facebook](#)

Language [English](#)

Also known as [Ma Trận](#)

Filming locations [Nashville, Tennessee, USA](#) (exterior scenes: skyline in opening Trinity rooftop chase)

Production companies [Warner Bros.](#) · [Village Roadshow Pictures](#) · [Groucho Film Partnership](#)

[See more company credits at IMDbPro](#)

Box office [Edit](#)

Budget \$63,000,000 (estimated)

Gross US & Canada \$172,076,928

Opening weekend US & Canada \$27,788,331 · Apr 4, 1999

Gross worldwide \$467,222,728

[Screenshot](#)

[Add to Watchlist](#)
Added by 1.0M users

max
PRIME VIDEO CHANNELS

Action **Sci-Fi**

When a beautiful stranger leads computer hacker Neo to a forbidding underworld, he discovers the shocking truth--the life he knows is the elaborate deception of an evil cyber-intelligence.

Directors [Lana Wachowski](#) · [Lilly Wachowski](#)

Writers [Lilly Wachowski](#) · [Lana Wachowski](#)

[Screenshot](#)

An example

- The IMDb movie database: Movies



Keanu Reeves

Biography

Jump to ▾

Overview

Born September 2, 1964 · Beirut, Lebanon

Birth name Keanu Charles Reeves

Nicknames The Wall · The One

Height 6' 1" (1.86 m)

Family

Children

No Children

Parents

Samuel Nowlin Reeves

[Patric Reeves](#)

Relatives

[Kim Reeves](#) (Sibling)

[Karina Miller](#) (Half Sibling)

Emma Reeves (Half Sibling)

Trademarks

Intense contemplative gaze

Deep husky voice

Known for playing stoic reserved characters

Friendly, down-to-earth personality



[The Matrix](#) (1999)

Edi

Full Cast & Crew

IMDbPro See agents for this cast & crew on IMDbPro [↗](#)

Directed by

[Lana Wachowski](#) ... (as The Wachowski Brothers)

[Lilly Wachowski](#) ... (as The Wachowski Brothers)

Writing Credits (WGA)

[Lilly Wachowski](#) ... (written by) (as The Wachowski Brothers) &

[Lana Wachowski](#) ... (written by) (as The Wachowski Brothers)

Cast (in credits order) verified as complete



[Keanu Reeves](#)

...

[Neo](#)



[Laurence Fishburne](#)

...

[Morpheus](#)



[Carrie-Anne Moss](#)

...

[Trinity](#)



[Hugo Weaving](#)

...

[Agent Smith](#)



[Gloria Foster](#)

...

[Oracle](#)



[Joe Pantoliano](#)

...

[Cypher](#)



[Marcus Chong](#)

...

[Tank](#)



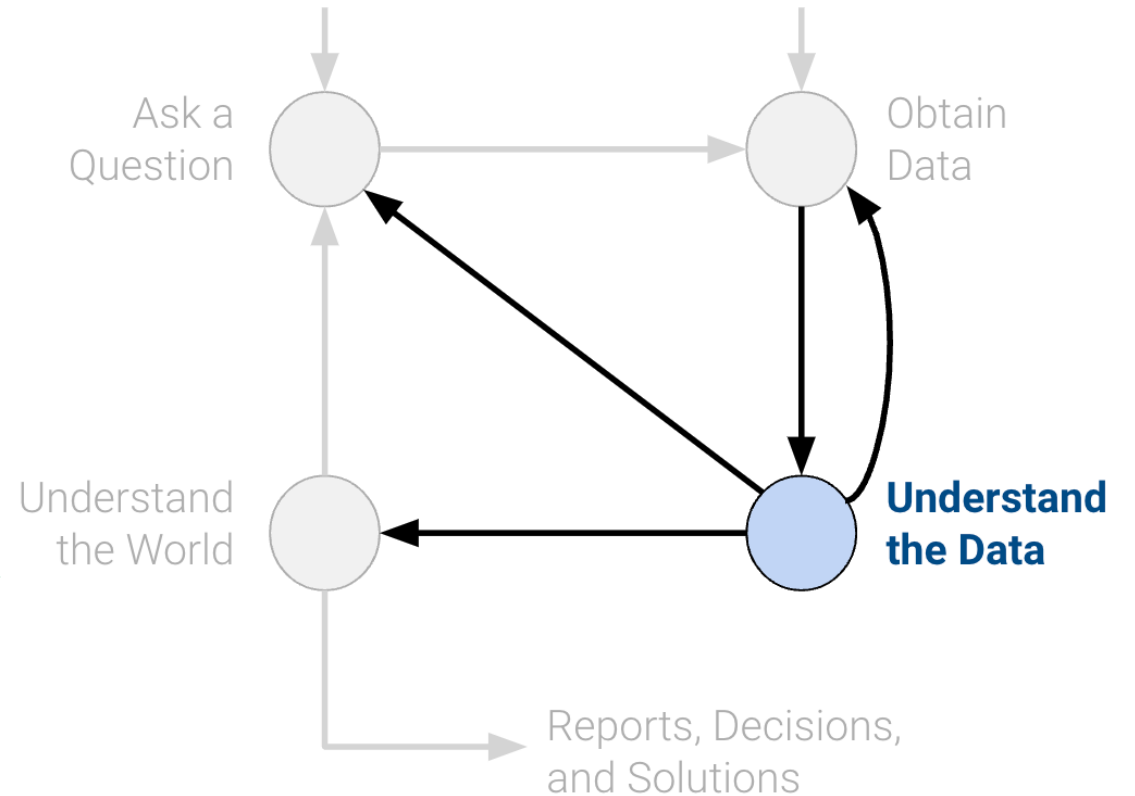
[Julian Arahanga](#)

...

[Apoc](#)

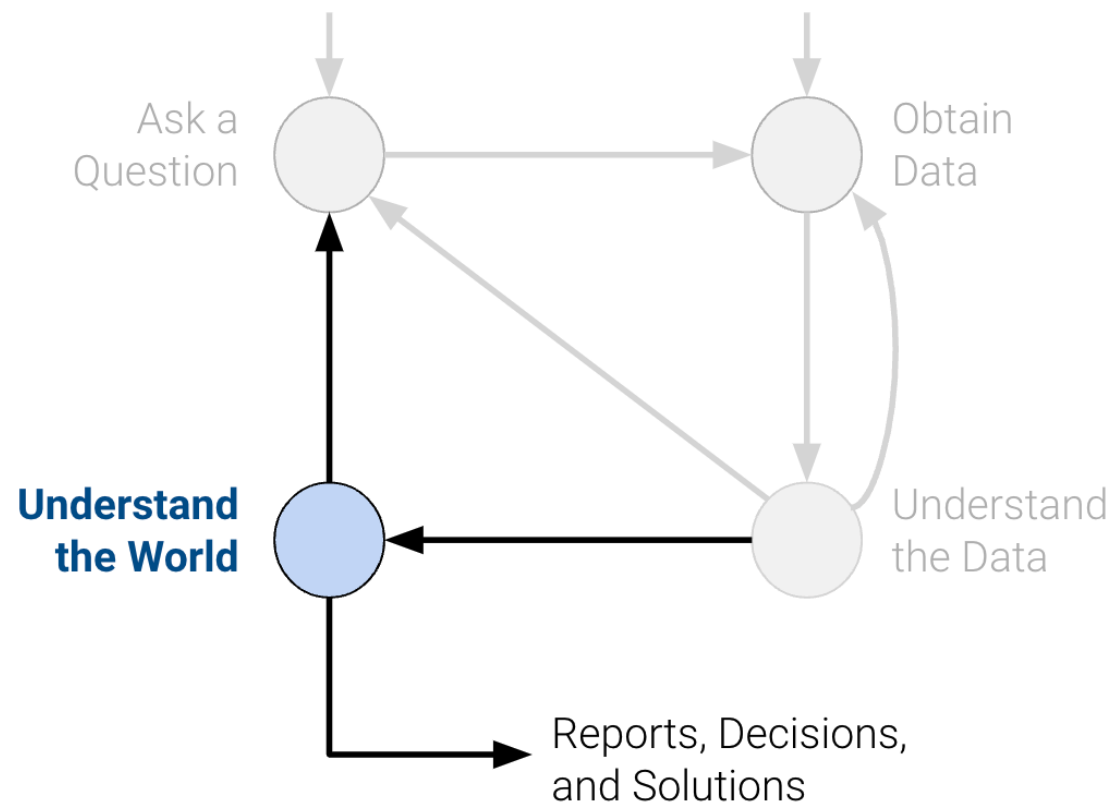
The data science life cycle

- How is our data organized and what does it contain?
- Do we already have relevant data?
- What are the biases, anomalies, or other issues with the data?
 - E.g., overrepresented by highly engaged user
- How do we transform the data to enable effective analysis?



The data science life cycle

- What does the data say about the world?
- Does it answer our questions or accurately solve the problem?
- How robust are our conclusions and can we trust the predictions?



Course Overview

Course Overview: Resources

<https://xinchenyu.github.io/csc380/>

CSC 380: Principle of Data Science

Overview

This course introduces students to principles of data science that are necessary for computer scientists to make effective decisions in their professional careers. A number of computer science sub-disciplines now rely on data collection and analysis. For example, computer systems are now complicated enough that comparing the execution performance of two different programs becomes a statistical estimation problem rather than a deterministic computation. This course teaches students the basic principles of how to properly collect and process data sources in order to derive appropriate conclusions from them. The course has main components of: basic probability, basic statistics and data wrangling, and basic data analysis using programming libraries.

Logistics info

Time and venue: Tuesday and Thursday 5:00-6:15pm at [M Pacheco ILC 130](#)

- [Syllabus](#)
- [Gradescope](#)
- [D2L course webpage](#)
- [Piazza link](#) (access code: `wildcats`)
- [Lecture participation self-report form](#)

Specific resources

- gradescope for assignment submission
- Piazza for discussions and Q&A.
- Readings and electronic textbooks
- Lecture slides

Most lecture accompanied by reading

- You are expected to read them

Attendance is required

Course Materials

Textbooks:

- Watkins, J., "An Introduction to the Science of Statistics: From Theory to Implementation"
- Wasserman, L. "All of Statistics: A Concise Course in Statistical Inference." Springer, 2004
- James, G., Witten, D., Hastie, T., & Tibshirani, R. An Introduction to Statistical Learning with Applications in Python. New York: Springer
- Steven S. Skiena, "The Data Science Design Manual", Springer, 2017
- Sam Lau, Joey Gonzalez, Deb Nolan, "Learning Data Science", O'Reilly, 2023

Lecture slides will be shared through course website.

Technologies/Libraries we will use



Expected Skills

- This class will use a fair amount of math
 - Probability and Statistics
 - Some basic Calculus and Linear Algebra
 - "Math should be there to aid understanding, not hindering it"-- Hal Daume III
- This class will require a fair amount of coding
 - Reading in / cleaning / visualizing data
 - Simulating random processes
 - Training and evaluating machine learning models
- Some assignment questions will be math, some will be coding

Course Overview

Course Objective *Introduction to basic concepts in data science and machine learning.*

Basic Probability	Basic Statistics	Data Handling and Visualization	Machine Learning
random events, variables, distributions, moments	descriptive statistics, estimation, hypothesis testing	reading, cleaning, preprocessing, visualization	predictive models, supervised learning, unsupervised learning

Tentative schedule

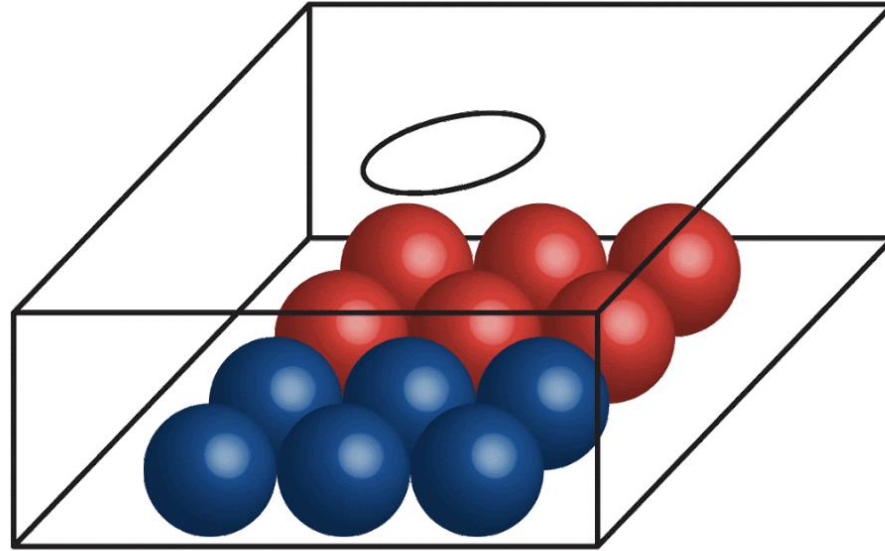
Note: All assignments due at 11:59 pm.

Dates	Topics	Assignments	Due Dates
Aug 25 Aug 27	Course Overview Basic Data Analysis 1	HW1 Out	
Sep 01 Sep 03	Probability 1.1 Probability 1.2		HW1 Due SEP 04
Sep 08 Sep 10	Probability 2.1 Probability 2.2	HW2 Out	
Sep 15 Sep 17	Probability 2.3 Probability 3.1	HW3 Out	HW2 Due SEP 16
Sep 22 Sep 24	Probability 3.2 Probability 3.3		
Sep 29 Oct 01	Probability 3.4 Probability 4.1	HW4 Out	HW3 Due SEP 29
Oct 06 Oct 08	Probability 4.2 Probability 4.3		
Oct 13 Oct 15	Midterm review 1 Midterm review 2		HW4 Due OCT 13
Oct 20 Oct 22	MIDTERM Basic Data Analysis 2.1	HW5 Out	

- Probability: 12 lectures
- Data analysis: 9 lectures
- Statistics: 4 lectures

Oct 27 Oct 29	Basic Data Analysis 2.2 Basic Data Analysis 3.1	Project Out
Nov 03 Nov 05	Basic Data Analysis 3.2 Basic Data Analysis 3.3	HW6 Out
Nov 10 Nov 12	Basic Data Analysis 3.4 Basic Data Analysis 4	
Nov 17 Nov 19	Basic Statistics 1.1 Basic Statistics 1.2	HW7 Out
Nov 24 Nov 26	Basic Statistics 2.1 No Class	
Dec 01 Dec 03	Basic Statistics 2.2 Basic Data Analysis 5	
Dec 08	Final review	
Dec 16	FINAL EXAM	

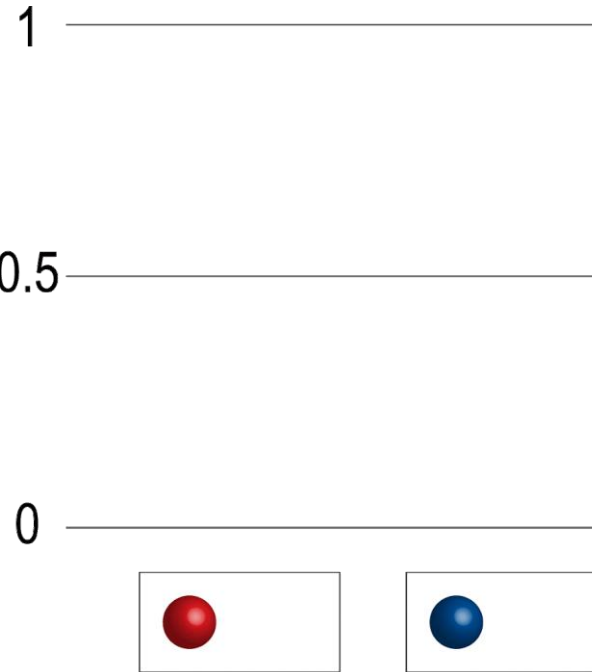
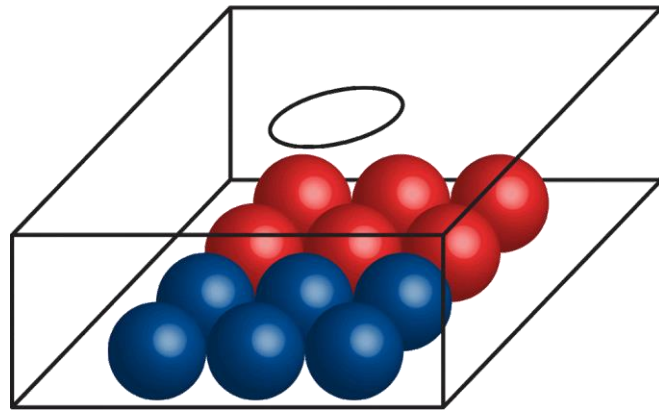
Probability



Probability is about **predicting chances using numbers**:

- Given that there are 6 red balls and 6 blue balls in the box.
- You pick 1 balls from the box. What is the probability that the ball is red?
- You pick 2 balls from the box. What is the probability that both balls are red?

Statistics



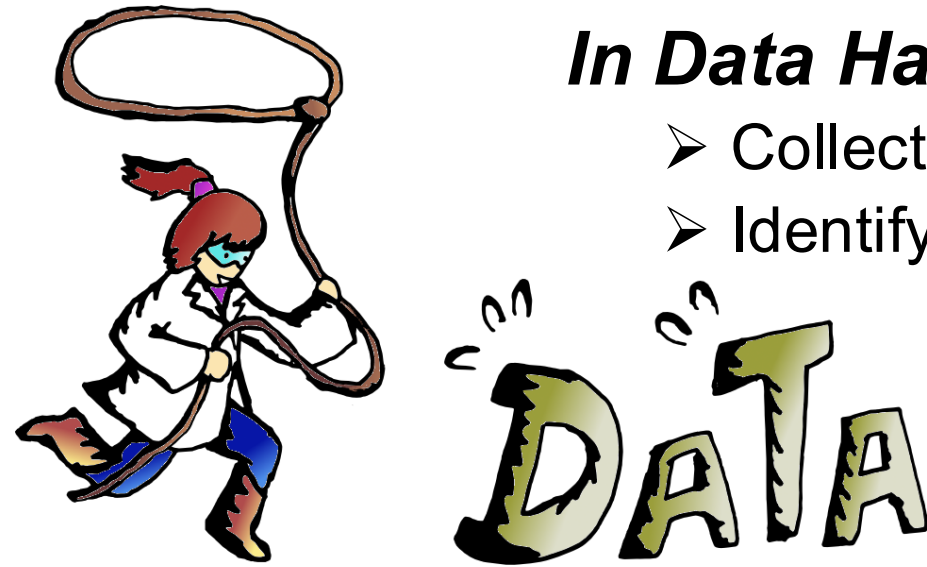
Statistics is about **turning raw data into insights**:

- There are 12 balls in the box, but you don't know how many are red.
- Experiment: pick 1 ball from the box each time then put it back, repeat 100 times.
- Suppose you observe 50 times are red, how many red balls would you guess? Why?

Data Handling and Visualization

In Data Handling we will learn to...

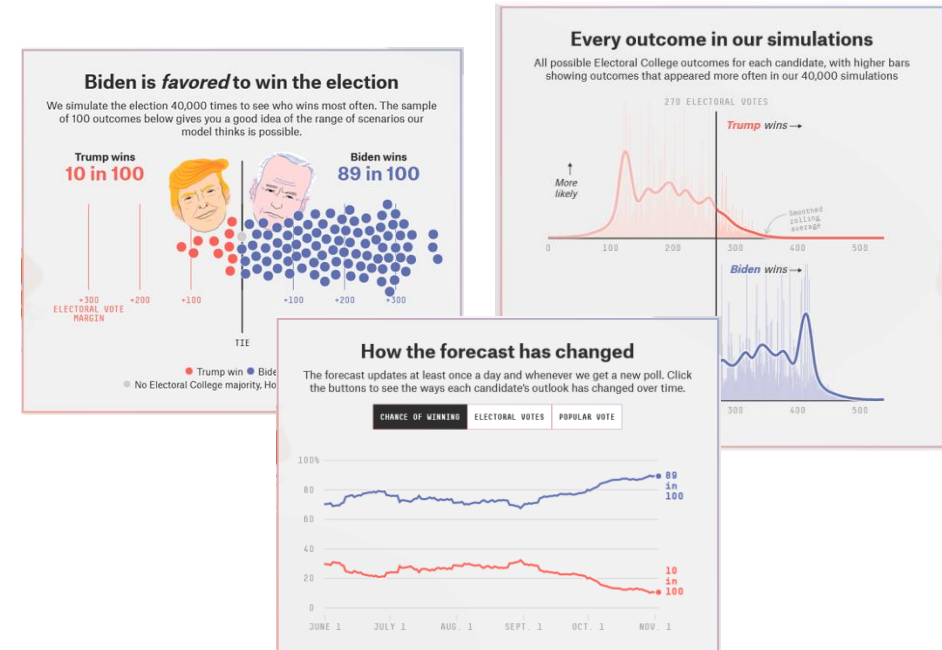
- Collect data
- Identify and avoid biased population samples
- Clean data and correct errors
- Transform and preprocess data (**wrangling**)



[Image Source: Code A Star]

In Data Visualization we will learn...

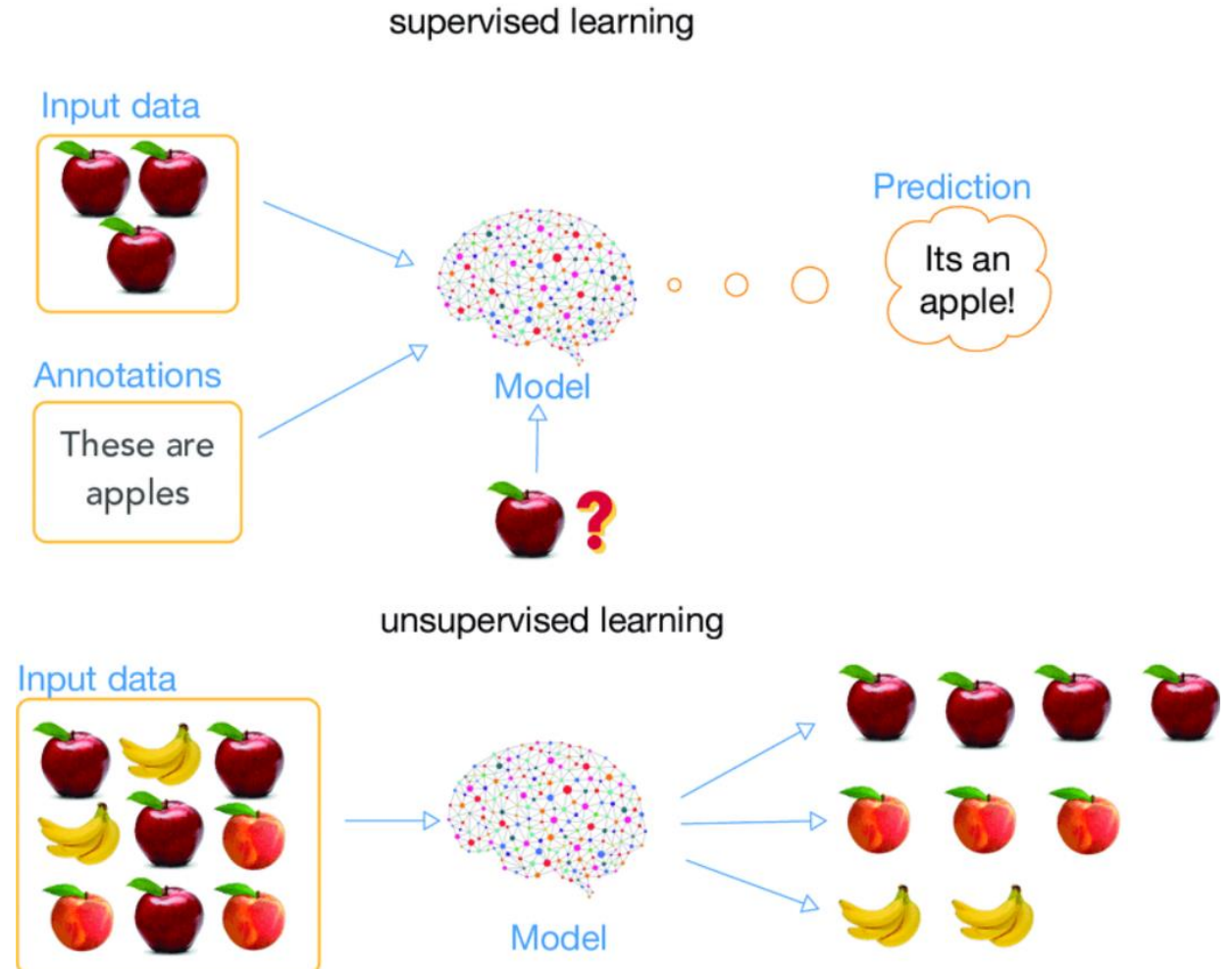
- Why visualization is important
- Exploratory data analysis
- Common forms of visualization
- Pitfalls and gotchas



Machine Learning

In Machine Learning we will learn...

- Principles of prediction
- Unsupervised vs. supervised learning
- Linear and nonlinear models
- Regression and classification



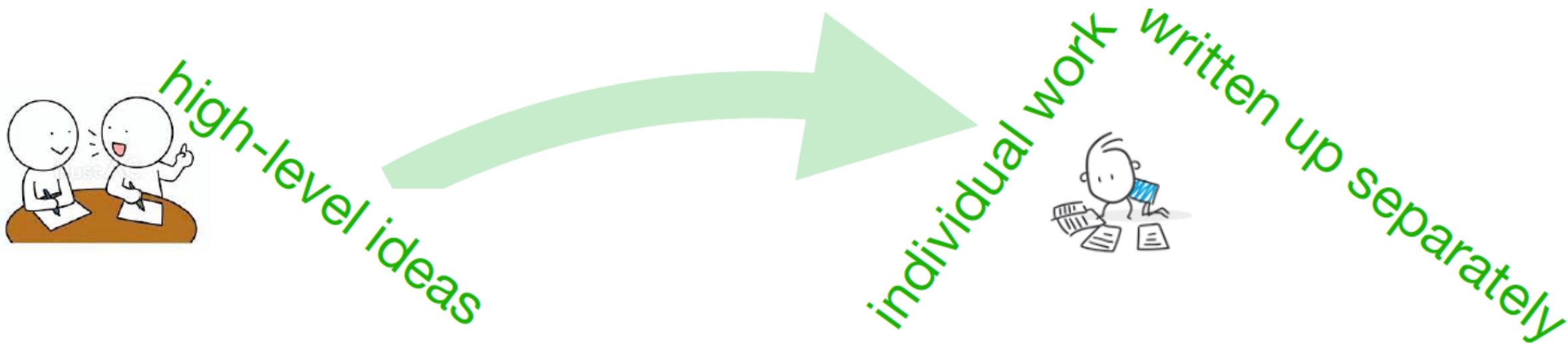
Communication

- Announcements will be made via [D2L](#)
- Homework submission: [Gradescope](#) (link: in course website)
- Outside-lecture communications will be through [Piazza](#) (signup code: in course website)
- The instructor and TAs can also be reached out through [email](#) for logistical or personal matters.
- Come to instructor or TAs [office hours](#).

Grading Items

- **Assignments (30% total, 5% each)**
 - Problem solving + coding, can be done in pairs
 - 7 assignments (lowest one dropped)
- **Quizzes (15% total, 1.5% each)**
 - 12 quizzes (lowest two dropped), typically on Tuesdays unless otherwise noted
- **Project (10%)**
 - Groups of size ≤ 3
- **Exams (45% total)**
 - 1 midterm (20%) + final (25%)

Homework policy



- If you discuss high-level ideas with a friend mention his/her name in your solution.
- Verbatim copying of solutions from any external source (web search, ChatGPT, etc.) is **not acceptable**. If you make use of any external source mention the source in your solution and make sure that your solution is your own work.
- Submissions after the due date and time are **not accepted**.

Bonus points

- **3% bonus:** All conditions below must be satisfied.
 - 3 “participation instances” in lectures
 - 3 “participation instances” in piazza
 - 3 appearances in instructor’s or TAs office hours

PAIR UP & DISCUSS



- “participation instance” in lecture: Answer a group (with neighbor) discussion question correctly
- “participation instance” in piazza: A correct answer to a technical question

Important Dates

- Sep 6: last date to self-withdraw without a 'W'
- Oct 20: midterm (on paper)
- Dec 11: project due
- Dec 16: final exam (on paper)

Mental Wellbeing

Some occasional stress / depression / anxiety is normal, but sometimes you may need extra help

- Non-emergency UA resources at Counseling & Psych Services Mon-Fri
 - Phone: 520-621-3334
 - Web: <https://health.arizona.edu/counseling-psych-services>
- Emergency resources in Tucson in this [Google Doc](#)

Inclusivity

We want to foster a comfortable and inclusive classroom experience

Please let me know if you feel excluded in any way, e.g.

- Improper use of pronouns
- Microaggressions
- Miscellaneous statements / interactions

You can message us on Piazza or discuss in person

Reading Assignments

- Robinson and Nolis, "What is Data Science?" (link from course schedule page)
- 'Probability and statistics cookbook' is a good cheat sheet. Download it from <http://statistics.zone/>